

Example 2–5. A 15-kVA, 2300/230-V transformer is to be tested to determine its excitation branch components, its series impedances, and its voltage regulation. The following test data have been taken from the transformer:

Open-circuit test (low voltage side)	Short-circuit test (high voltage side)
$V_{OC} = 230 \text{ V}$	$V_{SC} = 47 \text{ V}$
$I_{OC} = 2.1 \text{ A}$	$I_{SC} = 6.0 \text{ A}$
$P_{OC} = 50 \text{ W}$	$P_{SC} = 160 \text{ W}$

The data have been taken by using the connections shown in Figures 2–19 and 2–20.

- Find the equivalent circuit of this transformer referred to the high-voltage side.
- Find the equivalent circuit of this transformer referred to the low-voltage side.
- Calculate the full-load voltage regulation at 0.8 lagging power factor, 1.0 power factor, and at 0.8 leading power factor using the exact equation for V_p .
- Plot the voltage regulation as load is increased from no load to full load at power factors of 0.8 lagging, 1.0, and 0.8 leading.
- What is the efficiency of the transformer at full load with a power factor of 0.8 lagging?

Solution

- The turns ratio of this transformer is $a = 2300/230 = 10$. The excitation branch values of the transformer equivalent circuit referred to the secondary (low voltage) side can be calculated from the *open-circuit test* data, and the series elements referred to the primary (high voltage) side can be calculated from the *short-circuit test* data. From the open-circuit test data, the open-circuit impedance angle is

$$\theta_{OC} = \cos^{-1} \frac{P_{OC}}{V_{OC} I_{OC}}$$

$$\theta_{OC} = \cos^{-1} \frac{50 \text{ W}}{(230 \text{ V})(2.1 \text{ A})} = 84^\circ$$

The excitation admittance is thus

$$Y_E = \frac{I_{OC}}{V_{OC}} \angle -84^\circ$$

$$Y_E = \frac{2.1 \text{ A}}{230 \text{ V}} \angle -84^\circ \text{ S}$$

$$Y_E = 0.00913 \angle -84^\circ \text{ S} = 0.000954 - j0.00908 \text{ S}$$

The elements of the excitation branch referred to the secondary are

$$R_{C,S} = \frac{1}{0.000954} = 1050 \Omega$$

$$X_{M,S} = \frac{1}{0.00908} = 110 \Omega$$

From the short-circuit test data, the short-circuit impedance angle is

$$\theta_{SC} = \cos^{-1} \frac{P_{SC}}{V_{SC} I_{SC}}$$

$$\theta_{SC} = \cos^{-1} \frac{160 \text{ W}}{(47 \text{ V})(6 \text{ A})} = 55.4^\circ$$

The equivalent series impedance is thus

$$Z_{SE} = \frac{V_{SC}}{I_{SC}} \angle \theta_{SC}$$

$$Z_{SE} = \frac{47 \text{ V}}{6 \text{ A}} \angle 55.4^\circ \Omega$$

$$Z_{SE} = 7.833 \angle 55.4^\circ = 4.45 + j6.45 \Omega$$

The series elements referred to the primary side are

$$R_{eq,p} = 4.45 \, \Omega \quad X_{eq,p} = 6.45 \, \Omega$$

The resulting simplified equivalent circuit referred to the primary side can be found by converting the excitation branch values to the primary side.

$$R_{C,p} = a^2 R_{C,s} = (10)^2 (1050 \, \Omega) = 105 \, \text{k}\Omega$$

$$X_{M,p} = a^2 X_{M,s} = (10)^2 (110 \, \Omega) = 11 \, \text{k}\Omega$$

This equivalent circuit is shown in Figure 2-28a.

- (b) To find the equivalent circuit referred to the low-voltage side, it is simply necessary to divide the impedance by a^2 . Since $a = N_p/N_s = 10$, the resulting values are

$$R_C = 1050 \, \Omega \quad R_{eq} = 0.0445 \, \Omega$$

$$X_M = 110 \, \Omega \quad X_{eq} = 0.0645 \, \Omega$$

The resulting equivalent circuit is shown in Figure 2-28b.

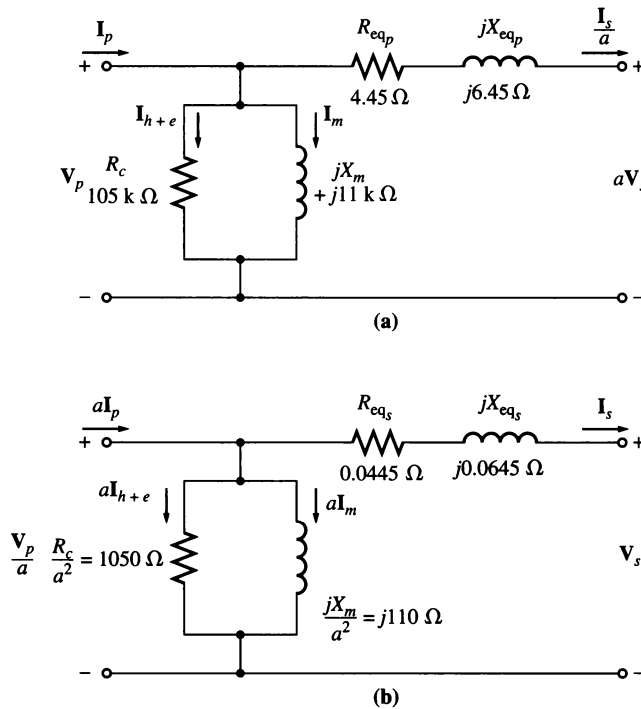


FIGURE 2-28

The transfer equivalent circuit for Example 2-5 referred to (a) its primary side and (b) its secondary side.

(c) The full-load current on the secondary side of this transformer is

$$I_{S,\text{rated}} = \frac{S_{\text{rated}}}{V_{S,\text{rated}}} = \frac{15,000 \text{ VA}}{230 \text{ V}} = 65.2 \text{ A}$$

To calculate V_P/a , use Equation (2-64):

$$\frac{V_P}{a} = V_S + R_{\text{eq}} I_S + jX_{\text{eq}} I_S \quad (2-64)$$

At PF = 0.8 lagging, current $I_S = 65.2 \angle -36.9^\circ \text{ A}$. Therefore,

$$\begin{aligned} \frac{V_P}{a} &= 230 \angle 0^\circ \text{ V} + (0.0445 \Omega)(65.2 \angle -36.9^\circ \text{ A}) + j(0.0645 \Omega)(65.2 \angle -36.9^\circ \text{ A}) \\ &= 230 \angle 0^\circ \text{ V} + 2.90 \angle -36.9^\circ \text{ V} + 4.21 \angle 53.1^\circ \text{ V} \\ &= 230 + 2.32 - j1.74 + 2.52 + j3.36 \\ &= 234.84 + j1.62 = 234.85 \angle 0.40^\circ \text{ V} \end{aligned}$$

The resulting voltage regulation is

$$\begin{aligned} \text{VR} &= \frac{V_P/a - V_{S,\text{fl}}}{V_{S,\text{fl}}} \times 100\% \\ &= \frac{234.85 \text{ V} - 230 \text{ V}}{230 \text{ V}} \times 100\% = 2.1\% \end{aligned} \quad (2-62)$$

At PF = 1.0, current $I_S = 65.2 \angle 0^\circ \text{ A}$. Therefore,

$$\begin{aligned} \frac{V_P}{a} &= 230 \angle 0^\circ \text{ V} + (0.0445 \Omega)(65.2 \angle 0^\circ \text{ A}) + j(0.0645 \Omega)(65.2 \angle 0^\circ \text{ A}) \\ &= 230 \angle 0^\circ \text{ V} + 2.90 \angle 0^\circ \text{ V} + 4.21 \angle 90^\circ \text{ V} \\ &= 230 + 2.90 + j4.21 \\ &= 232.9 + j4.21 = 232.94 \angle 1.04^\circ \text{ V} \end{aligned}$$

The resulting voltage regulation is

$$\text{VR} = \frac{232.94 \text{ V} - 230 \text{ V}}{230 \text{ V}} \times 100\% = 1.28\%$$

At PF = 0.8 leading, current $I_S = 65.2 \angle 36.9^\circ \text{ A}$. Therefore,

$$\begin{aligned} \frac{V_P}{a} &= 230 \angle 0^\circ \text{ V} + (0.0445 \Omega)(65.2 \angle 36.9^\circ \text{ A}) + j(0.0645 \Omega)(65.2 \angle 36.9^\circ \text{ A}) \\ &= 230 \angle 0^\circ \text{ V} + 2.90 \angle 36.9^\circ \text{ V} + 4.21 \angle 126.9^\circ \text{ V} \\ &= 230 + 2.32 + j1.74 - 2.52 + j3.36 \\ &= 229.80 + j5.10 = 229.85 \angle 1.27^\circ \text{ V} \end{aligned}$$

The resulting voltage regulation is

$$\text{VR} = \frac{229.85 \text{ V} - 230 \text{ V}}{230 \text{ V}} \times 100\% = -0.062\%$$

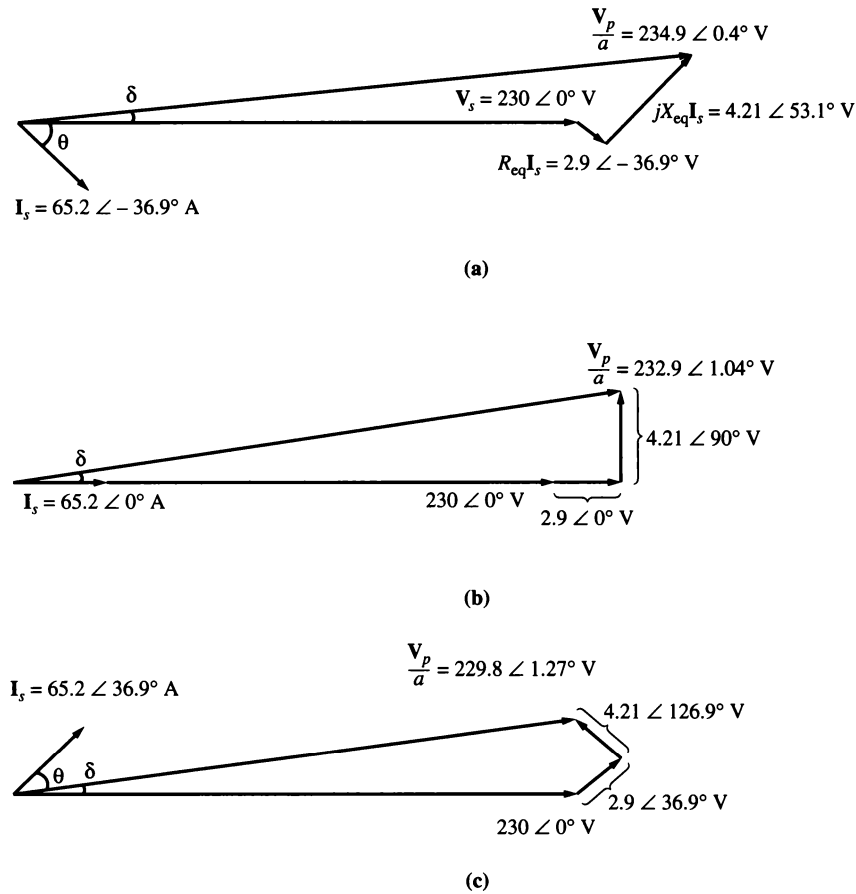


FIGURE 2-29
Transformer phasor diagrams for Example 2-5.

Each of these three phasor diagrams is shown in Figure 2-29.

- (d) The best way to plot the voltage regulation as a function of load is to repeat the calculations in part *c* for many different loads using MATLAB. A program to do this is shown below.

```
% M-file: trans_vr.m
% M-file to calculate and plot the voltage regulation
% of a transformer as a function of load for power
% factors of 0.8 lagging, 1.0, and 0.8 leading.
VS = 230; % Secondary voltage (V)
amps = 0:6.52:65.2; % Current values (A)
```

```

Req = 0.0445;           % Equivalent R (ohms)
Xeq = 0.0645;           % Equivalent X (ohms)

% Calculate the current values for the three
% power factors. The first row of I contains
% the lagging currents, the second row contains
% the unity currents, and the third row contains
% the leading currents.
I(1,:) = amps .* ( 0.8 - j*0.6);      % Lagging
I(2,:) = amps .* ( 1.0 );              % Unity
I(3,:) = amps .* ( 0.8 + j*0.6);      % Leading

% Calculate VP/a.
VPa = VS + Req.*I + j.*Xeq.*I;

% Calculate voltage regulation
VR = (abs(VPa) - VS) ./ VS .* 100;

% Plot the voltage regulation
plot(amps,VR(1,:), 'b-');
hold on;
plot(amps,VR(2,:), 'k-');
plot(amps,VR(3,:), 'r-.');
title ('Voltage Regulation Versus Load');
xlabel ('Load (A)');
ylabel ('Voltage Regulation (%)');
legend('0.8 PF lagging','1.0 PF','0.8 PF leading');
hold off;

```

The plot produced by this program is shown in Figure 2–30.

- (e) To find the efficiency of the transformer, first calculate its losses. The copper losses are

$$P_{\text{Cu}} = (I_S)^2 R_{\text{eq}} = (65.2 \text{ A})^2 (0.0445 \Omega) = 189 \text{ W}$$

The core losses are given by

$$P_{\text{core}} = \frac{(V_P/a)^2}{R_C} = \frac{(234.85 \text{ V})^2}{1050 \Omega} = 52.5 \text{ W}$$

The output power of the transformer at this power factor is

$$\begin{aligned}
 P_{\text{out}} &= V_S I_S \cos \theta \\
 &= (230 \text{ V})(65.2 \text{ A}) \cos 36.9^\circ = 12,000 \text{ W}
 \end{aligned}$$

Therefore, the efficiency of the transformer at this condition is

$$\begin{aligned}
 \eta &= \frac{V_S I_S \cos \theta}{P_{\text{Cu}} + P_{\text{core}} + V_S I_S \cos \theta} \times 100\% & (2-68) \\
 &= \frac{12,000 \text{ W}}{189 \text{ W} + 52.5 \text{ W} + 12,000 \text{ W}} \times 100\% \\
 &= 98.03\%
 \end{aligned}$$

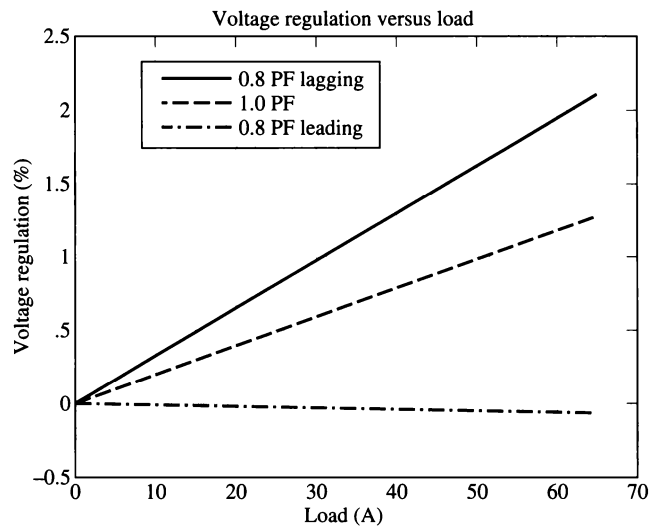


FIGURE 2-30
Plot of voltage regulation versus load for the transformer of Example 2-5.